WEAR-RAG: Improving Multi-Hop Retrieval-Augmented Generation via Weighted Evidence Aggregation

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***Abstract*—Many people now use retrieval-augmented generation (RAG) to ground enormous language model (LLM) outputs in outside evidence, therefore lowering hallucination and increasing factual trustworthiness. Three serious drawbacks of traditional RAG systems are, however: fixed-length chunking disturbs semantic coherence; single-query retrieval misses multi-hop reasoning chains; and equal treatment of all retrieved passages introduces low-quality or irrelevant evidence into the generation context. WEAR-RAG (Weighted Evidence Aggregation for Retrieval-Augmented Generation) is a modular, end-to-end pipeline addressing these issues with four integrated innovations: semantic chunking for topic-coherent document segmentation, rule-based query decomposition for multi-hop coverage, cross-encoder reranking for fine-grained relevancy estimate, and a new multi-signal evidence scoring framework. Combining dense retrieval similarity, cross-encoder relevance, and a content richness heuristic, the core contribution calculates a composite evidence score for every chunk. Before generation, chunks dropping below a quality threshold are filtered so that only high-value evidence enters the language model.**

We evaluate WEAR-RAG on 100 samples from the HotpotQA distractor validation split against a standard baseline RAG system. WEAR-RAG demonstrates consistent improvements: Exact Match rises from 0.0700 to 0.0800 (+14.29%), token-level F1 from 0.2136 to 0.2339 (+9.50%), ROUGE-L from 0.2123 to 0.2330 (+9.75%), BLEU-1 from 0.1582 to 0.1733 (+9.54%), and Retrieval Precision from 0.3480 to 0.3960 (+13.79%), while maintaining near-identical Mean Reciprocal Rank. These results highlight evidence curation as a compute-efficient alternative to scaling model size for improving RAG quality in multi-hop question answering.

***Index Terms*—Retrieval-Augmented Generation, Multi-hop Question Answering, Evidence Aggregation, Cross-Encoder Reranking, Semantic Chunking, HotpotQA.**

I. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation across tasks such as question answering, summarization, and dialogue. However, these models are inherently limited by the knowledge encoded during pre-training and remain susceptible to producing plausible but factually incorrect outputs—a phenomenon known as hallucination [1]. In high-stakes domains such as healthcare, legal reasoning, and enterprise knowledge management, such errors can have severe consequences. Retrieval-Augmented Generation (RAG) has emerged as a practical solution to this problem [2], coupling a retrieval mechanism with a generative model to ground outputs in externally retrieved evidence.

The standard RAG workflow involves three steps: (1) indexing a document corpus into a vector database for efficient search, (2) retrieving the top-k most similar passages for a given query, and (3) feeding these passages as context to the language model for answer generation.

A. Why Standard RAG Fails

Despite its conceptual elegance, the standard RAG paradigm introduces several practical limitations:

1) **Semantic Fragmentation:** Fixed-length chunking strategies can arbitrarily split semantically coherent passages across multiple chunks, degrading retrieval precision.
2) **Single-Query Limitation:** Complex multi-hop questions—such as "Which film has more awards, X or Y?"—require information from multiple sources that a single embedding query cannot simultaneously capture.
3) **Equal Evidence Treatment:** Standard top-k retrieval treats all returned passages as equally relevant, forwarding mixed-quality context to the generator without distinguishing highly relevant passages from noise.

B. Motivation: Evidence Selection as an Optimization Target

We observe that existing RAG systems invest significant effort in retrieval but treat evidence curation as an afterthought. The implicit assumption is that retrieval rank serves as a sufficient proxy for evidence quality. We challenge this assumption: context quality is an optimization target in its own right, not merely a byproduct of retrieval ranking. A system that explicitly scores, filters, and ranks evidence before generation can improve answer quality without modifying the underlying retrieval model.

C. Our Proposed Solution: WEAR-RAG

WEAR-RAG (Weighted Evidence Aggregation for RAG) introduces a multi-signal evidence scoring function that evaluates each retrieved chunk on three complementary dimensions—semantic similarity, cross-encoder relevance, and content richness—and retains only chunks exceeding a quality threshold. This evidence curation stage sits between retrieval and generation, acting as an intelligent filter that optimizes the input context provided to the language model. Unlike prior approaches that rely on implicit or single-signal ranking, WEAR-RAG explicitly models evidence quality as a multi-dimensional optimization problem, enabling fine-grained control over context construction before generation.

D. Contributions

1) **WEAR-RAG Architecture:** A complete, modular retrieval-to-generation pipeline integrating semantic chunking, query decomposition, cross-encoder reranking, and weighted evidence aggregation.
2) **Composite Evidence Scoring:** A novel, interpretable scoring function $S(c) = \alpha \cdot S_{\text{sim}} + \beta \cdot S_{\text{rank}} + \gamma \cdot S_{\text{dens}}$ that combines three complementary quality signals into a single evidence score.
3) **Ablation Study:** Systematic evaluation demonstrating that each pipeline component contributes positively and that their combination produces synergistic improvements.
4) **Empirical Validation:** Consistent improvements over baseline RAG across six evaluation metrics on the HotpotQA benchmark.
5) **Deployable System:** Full implementation with a web interface, mock testing mode, and visualization tools for evidence inspection.

Unlike prior approaches that rely on implicit ranking signals, WEAR-RAG explicitly formulates evidence selection as a multi-dimensional optimization problem.

II. Background

This section provides the technical foundations underlying WEAR-RAG. We cover the core concepts that our system builds upon, establishing the vocabulary and mathematical framework used throughout the paper. Modern LLMs such as GPT [14], LLaMA [22], and Mistral [23] have advanced language understanding significantly, yet remain constrained by parametric knowledge boundaries.

A. Retrieval-Augmented Generation

RAG systems augment language model generation with retrieved external knowledge. Given a query q, a retriever $\mathcal{R}$ fetches a set of relevant passages $\{p_1, \ldots, p_k\}$ from a corpus $\mathcal{C}$. These passages are concatenated with the query to form a context $C = [q; p_1; \ldots; p_k]$, which is passed to a generator $\mathcal{G}$ to produce the answer:

$$a = \mathcal{G}(q, \mathcal{R}(q, \mathcal{C})) \tag{1}$$

The quality of the generated answer a is fundamentally bounded by the quality of the retrieved passages. If the retriever returns irrelevant or noisy passages, the generator receives corrupted context, leading to hallucinated or incomplete answers. This garbage-in, garbage-out property motivates the need for intermediate evidence curation between retrieval and generation.

B. Embeddings and Vector Search

Dense retrieval relies on embedding both queries and passages into a shared high-dimensional vector space. A bi-encoder model f_θ maps text to d-dimensional vectors:

$$\mathbf{v}_q = f_\theta(q), \quad \mathbf{v}_p = f_\theta(p), \quad \mathbf{v}_q, \mathbf{v}_p \in \mathbb{R}^d \tag{2}$$

Relevance is computed as the cosine similarity between query and passage embeddings:

$$\text{sim}(q, p) = \frac{\mathbf{v}_q^\top \mathbf{v}_p}{\|\mathbf{v}_q\| \cdot \|\mathbf{v}_p\|} \tag{3}$$

When embeddings are L2-normalized (as in WEAR-RAG), cosine similarity reduces to inner product, enabling exact nearest-neighbor search via FAISS IndexFlatIP. FAISS [5] provides highly optimized implementations for billion-scale similarity search. The key limitation of bi-encoders is the independence assumption: queries and passages are encoded separately, so the model cannot attend to fine-grained interactions between query terms and passage terms during encoding.

C. Cross-Encoder Reranking

Cross-encoders address the bi-encoder limitation by jointly processing the query-passage pair through a single transformer:

$$\text{relevance}(q, p) = \sigma\big(\text{CrossEncoder}([q; p])\big) \tag{4}$$

where σ is the sigmoid function mapping raw logits to $[0, 1]$. Because the cross-encoder attends to all token-to-token interactions between q and p, it captures subtleties such as negation, coreference, and entity disambiguation that bi-encoders miss. The tradeoff is computational cost: cross-encoders require $O(k)$ forward passes for k candidates, compared to a single embedding lookup for bi-encoders. This motivates a two-stage pipeline: fast bi-encoder retrieval followed by slower cross-encoder reranking [6].

D. Multi-Hop Reasoning

Multi-hop questions require reasoning across multiple pieces of evidence, often from different documents. For example, answering "What is the birth year of the director of the 2007 Golden Bear winner?" requires: (1) identifying which film won the 2007 Golden Bear, then (2) finding the birth year of that film's director. A single retrieval query cannot simultaneously capture both information needs.

Question decomposition addresses this by breaking a complex query into simpler sub-queries, each targeting one reasoning step. The sub-queries are answered independently, and their evidence is aggregated to produce the final answer. HotpotQA [7] is the standard benchmark for evaluating multi-hop QA systems, containing questions that require reasoning over two supporting documents amid distractor paragraphs.

E. Document Chunking Strategies

Chunking is the process of partitioning documents into retrieval-sized units. The choice of chunking strategy directly affects retrieval quality:

Fixed-length chunking splits documents at regular intervals (e.g., every 300 words or 512 tokens). It is simple and deterministic but ignores semantic boundaries. A topic discussion spanning 400 words will be split across two chunks, with each half lacking full context—degrading both retrieval precision and generation quality.

Sentence-level chunking uses individual sentences as retrieval units. While respecting natural language boundaries, single sentences often lack sufficient context for meaningful retrieval. A pronoun like "he" requires its antecedent from a previous sentence, which may be in a different chunk.

Semantic chunking (used in WEAR-RAG) detects topic boundaries by monitoring embedding similarity between adjacent sentences. Sentence embeddings are computed using models based on the Sentence-BERT framework [15]. When similarity drops below a threshold, a new chunk begins. This produces variable-length, topic-coherent units with optional sentence overlap for continuity.

Recursive chunking hierarchically splits documents at natural boundaries (paragraphs, sections, sentences) until chunks meet size constraints. This respects document structure but does not consider semantic similarity between adjacent segments.

III. Related Work

A. Standard RAG Systems

The RAG paradigm was formalized by Lewis et al. [2], introducing two variants: RAG-Sequence, which conditions the entire generation on a single retrieved document, and RAG-Token, which allows different tokens to attend to different documents. REALM [3] pre-trains the retriever end-to-end with the language model, treating retrieval as a latent variable. However, most production RAG systems adopt a simpler decoupled pipeline where retriever and generator are independently optimized. This decoupling creates an opportunity for intermediate processing—such as evidence aggregation—that WEAR-RAG exploits.

B. Hybrid and Dense Retrieval

Dense Passage Retrieval (DPR) [4] replaced sparse lexical matching methods (BM25 [16], TF-IDF) with dense bi-encoder representations, enabling semantic similarity-based retrieval that captures meaning beyond surface-level word overlap. Bi-encoders embed queries and passages independently into a shared vector space, allowing efficient approximate nearest-neighbor search. Hybrid approaches combine sparse and dense signals for improved robustness on keyword-heavy queries. WEAR-RAG uses pure dense retrieval but compensates for its limitations through downstream cross-encoder reranking and multi-signal evidence aggregation.

C. Reranking Methods

Cross-encoder reranking has emerged as a powerful technique for refining retrieval results. Nogueira and Cho [6] demonstrated that BERT-based [13] cross-encoder reranking significantly improves retrieval quality in a two-stage pipeline. ColBERT [17] introduced late interaction patterns for more efficient reranking. More recently, models like BGE-reranker and MonoT5 have shown strong performance across diverse benchmarks. WEAR-RAG adopts full cross-encoder reranking but uniquely extends it: rather than simply selecting the reranked top-k, it incorporates reranker scores as one component of a multi-signal evidence scoring function, enabling more nuanced evidence selection.

D. Multi-Hop QA

Multi-hop question answering requires reasoning across multiple evidence passages. Benchmarks like HotpotQA [7], MuSiQue [19], and 2WikiMultiHopQA evaluate this capability with questions requiring exactly two or more supporting facts. Question decomposition [8] breaks complex queries into simpler, independently answerable sub-questions. Iterative retrievers like MDR [18] perform sequential retrieval rounds where each iteration is conditioned on passages retrieved so far. WEAR-RAG takes a different approach: explicit decomposition with parallel (not sequential) retrieval, followed by evidence aggregation that unifies results from all sub-queries.

E. Evidence Selection and Filtering

A growing body of work recognizes that not all retrieved passages deserve equal representation in the generation context. Izacard and Grave [9] demonstrated that Fusion-in-Decoder (FiD) benefits from processing many passages in parallel, but even FiD degrades with noisy context. RECOMP [11] compresses retrieved passages into shorter summaries before generation. Citation-grounded prompting [10] forces the generator to attribute claims to specific sources, improving verifiability. WEAR-RAG takes a complementary approach: rather than compressing individual passages, it scores entire chunks on multiple quality dimensions and retains only those exceeding a quality threshold. Table I systematically positions WEAR-RAG against these prior approaches.

In contrast to these approaches, WEAR-RAG focuses on explicit, interpretable evidence scoring prior to generation, rather than implicit ranking or post-hoc filtering. Recent work on Self-RAG [20] introduces self-reflective retrieval where the model decides when to retrieve and evaluates its own outputs; WEAR-RAG complements this direction by focusing on evidence quality *before* generation rather than post-generation reflection. WEAR-RAG's key differentiators are: (1) semantic rather than fixed chunking, (2) explicit query decomposition, (3) multi-signal evidence scoring, and (4) full interpretability through score decomposition.

IV. Problem Formulation

We formalize the WEAR-RAG problem as follows.

TABLE I
SYSTEMATIC COMPARISON OF WEAR-RAG WITH PRIOR RAG APPROACHES

System	Chunking	Retriever	Reranker	Decomp.	Evidence Sel.	Interpret.
RAG [2]	Fixed	DPR	None	None	Top-k	Low
REALM [3]	Fixed	Learned	None	None	Top-k	Low
FiD [9]	Fixed	DPR	None	None	All-k fusion	Low
MDR [18]	Fixed	Iterative	None	Implicit	Top-k	Medium
RECOMP [11]	Fixed	DPR	None	None	Compression	Medium
WEAR-RAG	**Semantic**	**Dense**	**Cross-Enc.**	**Explicit**	**Weighted**	**High**

A. Definitions

Query. Let q denote an input question in natural language. For multi-hop questions, q implicitly encodes multiple information needs.

Document Corpus. Let $D = \{d_1, d_2, \ldots, d_n\}$ be a set of n documents. Each document d_i is a variable-length text passage.

Chunks. Chunking partitions each document into smaller units. Let $C = \{c_1, c_2, \ldots, c_N\}$ be the complete set of chunks derived from D, where $N \gg n$ since each document produces multiple chunks.

Evidence Set. An evidence set $E \subseteq C$ is a subset of chunks selected to support answer generation. Each evidence item $e_i \in E$ carries a quality score $S(e_i) \in [0, 1]$.

Answer. The generated answer a is a natural language string produced by a language model conditioned on q and E.

B. Retrieval

Retrieval is the process of selecting candidate chunks from C given a query q:

$$\mathcal{R}(q, C) = \{c_i \in C : \text{sim}(q, c_i) \geq \text{threshold}\} \tag{5}$$

In practice, retrieval returns the top-k chunks ranked by embedding similarity. For multi-hop queries, WEAR-RAG decomposes q into sub-queries $Q = \{q_1, \ldots, q_m\}$ and retrieves candidates per sub-query.

C. Evidence Scoring

Evidence scoring assigns a composite quality score to each candidate chunk:

$$S(c) = \alpha \cdot S_{\text{sim}}(c) + \beta \cdot S_{\text{rank}}(c) + \gamma \cdot S_{\text{dens}}(c) \tag{6}$$

where $\alpha + \beta + \gamma = 1$ ensures the score is in $[0, 1]$.

D. Objective

The WEAR-RAG objective is to find the evidence set E^* that maximizes answer quality:

$$E^* = \underset{E \subseteq C, |E| \leq B}{\arg\max} \ \text{Quality}\big(\mathcal{G}(q, E)\big) \tag{7}$$

subject to $S(e_i) \geq \theta \ \ \forall e_i \in E$ and $\sum_{e_i \in E} |e_i| \leq B$ (token budget constraint). Here, θ is the quality threshold and B is the maximum context tokens.

V. PROPOSED METHOD: WEAR-RAG

A. Pipeline Overview

WEAR-RAG executes a six-stage pipeline, as illustrated in Fig. 1:

1) **Semantic Chunking:** Split documents into topic-coherent chunks.
2) **Query Decomposition:** Expand q into sub-queries $Q = \{q_1, \ldots, q_m\}$.
3) **Dense Retrieval:** Retrieve candidates per sub-query from FAISS.
4) **Cross-Encoder Reranking:** Re-score query-chunk pairs.
5) **Weighted Evidence Aggregation:** Compute composite scores, filter, rank.
6) **Grounded Generation:** Build evidence-constrained prompt and generate.

B. Semantic Chunking

Unlike fixed-size chunking which can split mid-topic (Fig. 2), WEAR-RAG detects topic boundaries by monitoring embedding similarity between adjacent sentences. Each document is sentence-tokenized using NLTK [24] and embedded using the BAAI/bge-small-en model. Adjacent sentence cosine similarity is computed as:

$$\text{sim}(i, i{+}1) = \frac{\mathbf{s}_i^\top \mathbf{s}_{i+1}}{\|\mathbf{s}_i\| \cdot \|\mathbf{s}_{i+1}\|} \tag{8}$$

A new chunk boundary is inserted when $\text{sim}(i, i{+}1) < \tau$ (topic shift) or when the chunk exceeds the maximum size. Minimum chunk size is enforced to avoid degenerate micro-chunks, and 1-sentence overlap at boundaries preserves cross-chunk continuity.

TABLE II
SEMANTIC CHUNKING CONFIGURATION

Parameter	Value
Similarity threshold (τ)	0.75
Minimum chunk size	50 tokens
Maximum chunk size	300 tokens
Sentence overlap	1 sentence

C. Query Decomposition

Multi-hop questions require information about multiple entities or concepts. WEAR-RAG's rule-based decomposer handles three primary patterns:

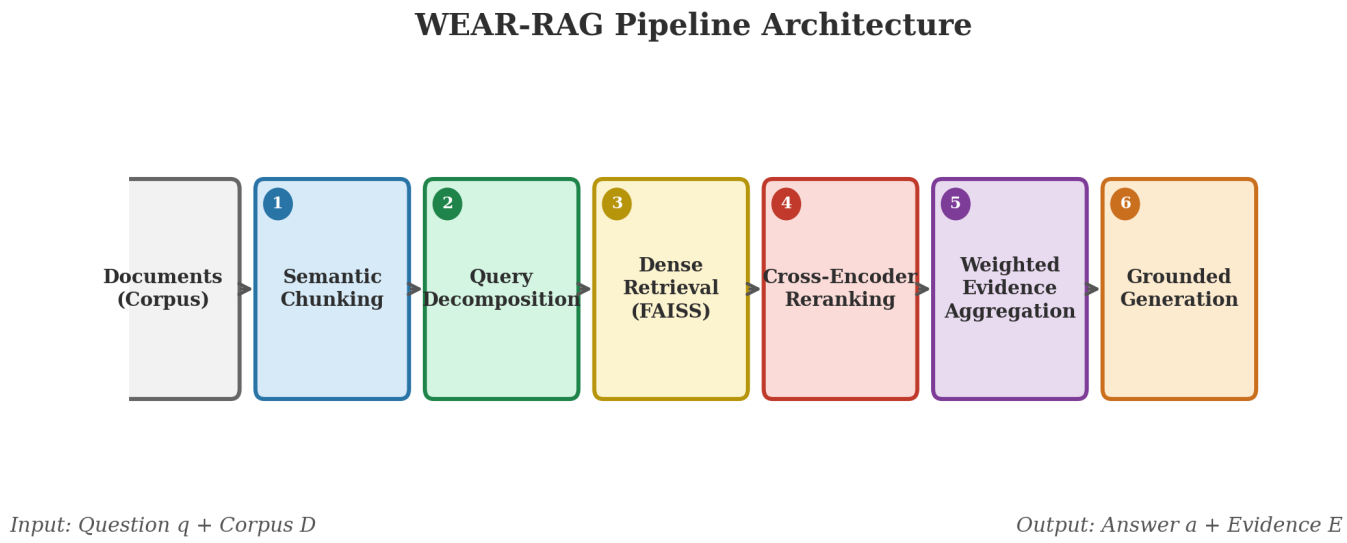

Fig. 1. WEAR-RAG pipeline architecture showing the six processing stages from document ingestion to grounded answer generation. Stage 5 (Weighted Evidence Aggregation) is the core contribution.

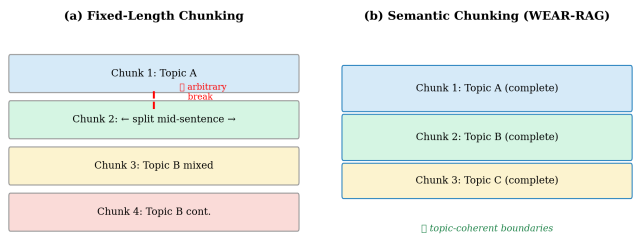

Fig. 2. (a) Fixed-length chunking splits mid-topic. (b) Semantic chunking preserves topic coherence within each chunk.

- **Comparison:** "Which is better, X or Y?" $\rightarrow$ separate sub-queries for X and Y.
- **Causal:** "Why does X happen?" $\rightarrow$ sub-queries for X's definition and causes.
- **Compound:** "What is X and how does Y relate?" $\rightarrow$ separate sub-queries for X and Y.

The original query is always retained to preserve intent, with a maximum of 4 sub-queries to prevent retrieval dilution. An optional LLM-based decomposer provides richer decomposition with rule-based fallback for reliability.

D. Two-Stage Retrieval and Reranking

Stage 1 — Dense Retrieval: For each sub-query, the top-20 candidates are retrieved using BAAI/bge-small-en embeddings via FAISS IndexFlatIP.

Stage 2 — Cross-Encoder Reranking: Retrieved candidates are re-scored using BAAI/bge-reranker-base. Raw logits are mapped via sigmoid to produce interpretable relevance scores:

$$S_{\text{rank}}(q, c) = \sigma\big(\text{CrossEncoder}(q, c)\big) = \frac{1}{1 + e^{-\text{logit}(q,c)}} \tag{9}$$

Multi-query reranking deduplicates across sub-queries, retaining the highest score per unique chunk.

E. Multi-Signal Evidence Scoring (Core Contribution)

This is the central innovation of WEAR-RAG. Rather than relying on a single retrieval score, each candidate chunk receives a composite evidence score integrating three complementary quality signals:

$$S(c) = \alpha \cdot S_{\text{sim}}(c) + \beta \cdot S_{\text{rank}}(c) + \gamma \cdot S_{\text{rich}}(c) \tag{10}$$

This formulation is intentionally designed as an extensible framework: additional quality signals (e.g., temporal recency, source authority, or learned features) can be incorporated by extending the weight vector, and the fixed weights can be replaced with learned or adaptive weighting schemes without modifying the pipeline architecture.

Similarity Score S_{sim}: The cosine similarity between the query embedding and the chunk embedding, as computed by FAISS. This captures broad semantic relevance—how topically related the chunk is to the query. Range: $[0, 1]$.

Reranker Score S_{rank}: The sigmoid-normalized cross-encoder output, capturing fine-grained query-passage interactions including entity matching, negation handling, and contextual relevance. Range: $[0, 1]$.

Content Richness Score S_{rich}: A lightweight heuristic estimating how information-dense a chunk is:

$$S_{\text{rich}} = 0.5 \cdot L_s + 0.3 \cdot \text{TTR} + 0.2 \cdot E_d \tag{11}$$

where:

- L_s (Length Factor): $\min\big(1.0,\ \log(1+|w|)/\log(1+200)\big)$, rewarding substantive passages.
- TTR (Type-Token Ratio): ratio of unique tokens to total tokens, measuring vocabulary diversity.
- E_d (Entity Density): fraction of tokens with uppercase initials, serving as a proxy for named entities and technical terms.

While simple, this heuristic captures structural signals correlated with information density. We intentionally favor lightweight, interpretable features over learned signals to maintain efficiency and transparency.

Weight Constraint: $\alpha + \beta + \gamma = 1.0$. Default values are $\alpha = 0.5$, $\beta = 0.4$, $\gamma = 0.1$, reflecting the intuition that semantic similarity provides the broadest signal, cross-encoder reranking provides the most precise signal, and content richness serves as a complementary quality indicator.

Threshold Filtering: Chunks are retained only if $S(c) \geq \theta$ with $\theta = 0.3$. Surviving chunks are sorted by score and trimmed to fit within the LLM's token budget (Fig. 3).

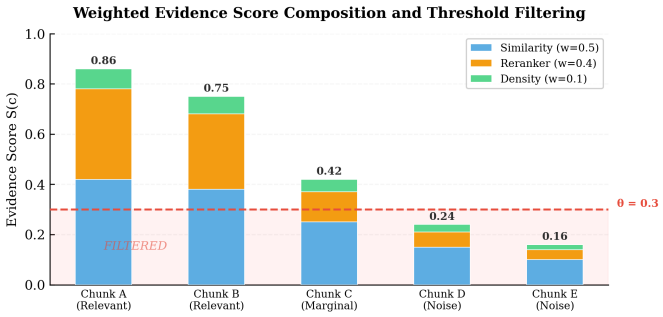

Fig. 3. Weighted evidence score composition for five candidate chunks. Stacked bars show the contribution of each component. The dashed red line indicates the quality threshold ($\theta = 0.3$). Chunks D and E are filtered before generation.

F. Grounded Generation

The generator receives labeled evidence blocks with source attribution and strict instructions:

- Answer using *only* the provided evidence.
- If evidence is insufficient, output "I don't know" rather than fabricating.
- Cite evidence block numbers in the response.

The default backend is Ollama-hosted Mistral 7B. A deterministic mock generator is provided for testing without heavy model dependencies.

VI. System Architecture

A. Module Design

WEAR-RAG is implemented as a fully modular Python system with clear separation of concerns, as shown in Fig. 4. Each component can be independently tested, replaced, or extended.

TABLE III
WEAR-RAG Component Summary

Component	Responsibility
Config. Manager	Centralized hyperparameter dataclasses
Doc. Processor	Semantic chunking with topic detection
Embedding Engine	Dense bi-encoder embeddings (BGE)
Vector Store	FAISS indexing, search, persistence
Query Decomposer	Rule-based and LLM decomposition
Reranker	Cross-encoder relevance refinement
Evidence Aggregator	Weighted scoring, filtering, budgeting
LLM Generator	Ollama generation and mock mode
Evaluator	Metric computation and CSV export
Pipeline Orchestrator	End-to-end coordination and CLI
Web API	Flask-based REST endpoints

B. Data Flow

The data flow through WEAR-RAG follows a strictly linear pipeline:

Document Ingestion: Raw documents → Sentence tokenization → Embedding computation → Semantic chunk boundaries → Chunk objects with metadata.

Query Processing: User question → Query decomposition → Sub-query list → Per-sub-query FAISS search → Candidate chunk sets.

Evidence Curation: Candidates → Cross-encoder reranking → Score computation → Threshold filtering → Token budget trimming → Ordered evidence set.

Answer Generation: Evidence set → Prompt construction → LLM inference → Grounded answer with citations.

C. Implementation Details

A **Shared Model Registry** loads heavy models (BAAI/bge-small-en, BAAI/bge-reranker-base) exactly once and shares them across pipeline variants, reducing evaluation time by $\approx 3\times$. The registry uses lazy-loading properties to defer initialization until first access. Per-sample temporary vector indices prevent information leakage across evaluation questions while heavy models remain loaded in memory.

D. Computational Complexity

Table IV summarizes the computational cost of each pipeline stage. The primary overhead versus baseline RAG is the cross-encoder reranking stage, a deliberate tradeoff that trades additional compute for higher evidence quality. WEAR-RAG intentionally trades additional reranking compute for improved context quality, prioritizing answer accuracy over minimal latency.

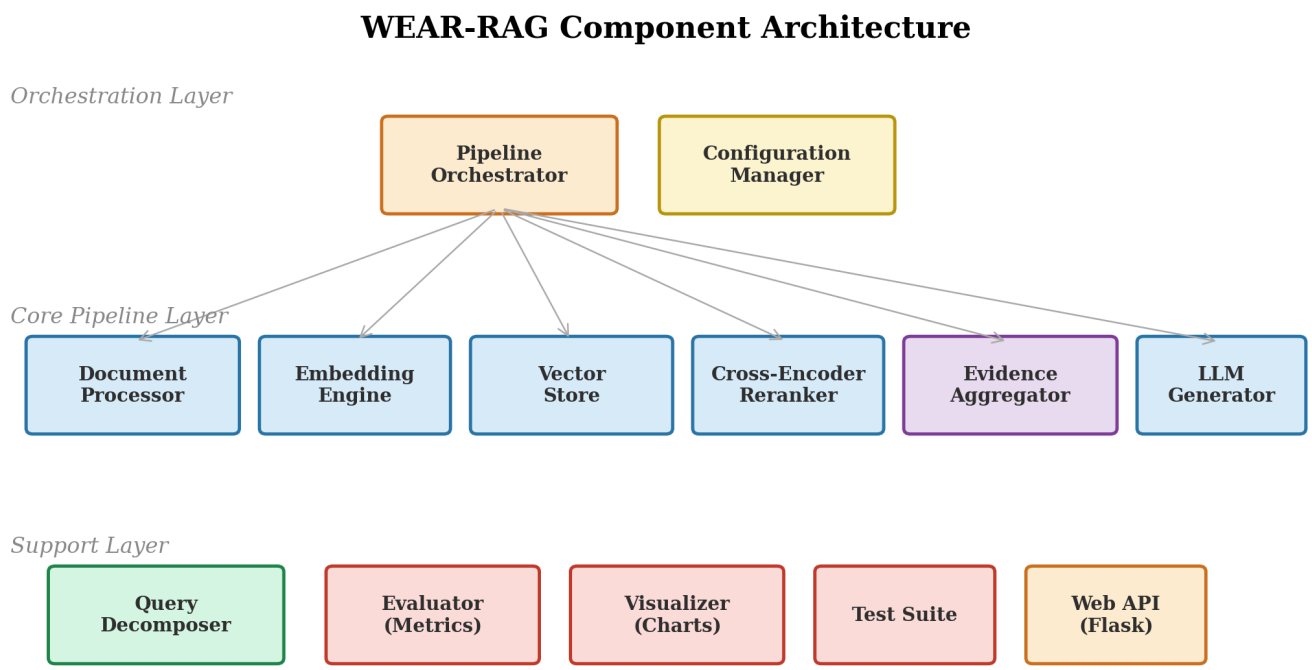

Fig. 4. WEAR-RAG component architecture showing three layers: Orchestration (pipeline controller, configuration manager), Core Pipeline (document processor, embedding engine, vector store, reranker, evidence aggregator, generator), and Support (query decomposer, evaluator, visualizer, test suite, web API).

TABLE IV
COMPUTATIONAL COMPLEXITY BY PIPELINE STAGE

Stage	Complexity
Semantic chunking	$O(n)$ per document
Embedding computation	$O(N \cdot d)$
Dense retrieval (FAISS)	$O(N \cdot d)$ per query
Cross-encoder reranking	$O(m \cdot k)$ forward passes
Evidence aggregation	$O(m \cdot k)$, negligible
Generation	O(context length)

where N is the total number of chunks, d is the embedding dimension (384), m is the number of sub-queries (up to 4), and k is the retrieval depth per sub-query (20). The reranking stage requires $m \times k \leq 80$ cross-encoder forward passes, taking approximately 2–5 seconds on a consumer GPU. For latency-sensitive deployments, this can be reduced by lowering m or k.

E. Memory Optimization

The Model Registry pattern ensures that embedder and reranker models (∼200 MB combined) are loaded once and shared across all pipeline variants during evaluation. Without this optimization, evaluating three pipeline variants would require loading these models three times, nearly tripling memory usage and initialization time.

F. Web Application

A Flask-based application provides interactive QA capabilities through three REST endpoints:

- **Health Check:** Returns model readiness status, useful for deployment monitoring and load balancer integration.
- **Document Upload:** Accepts TXT and PDF files, processes them through the semantic chunker, and adds them to the vector store. Supports incremental document ingestion.
- **Question Answering:** Takes a question string, runs the full WEAR-RAG pipeline, and returns the answer along with evidence items including scores and source identifiers.

The web interface was instrumental during development for human-in-the-loop error analysis and qualitative inspection of evidence rankings. Both production (Ollama) and mock testing modes are supported.

VII. EXPERIMENTAL SETUP

A. Dataset

Experiments use the HotpotQA benchmark [7], specifically the distractor validation split. HotpotQA requires multi-hop reasoning across multiple supporting documents and includes distractor paragraphs. We evaluate on 100 samples to balance reproducibility with local compute constraints. This subset is sufficient to observe consistent performance trends while maintaining feasibility under local compute constraints.

B. Compared Systems

TABLE V
SYSTEM COMPARISON CONFIGURATION

System	Chunk.	Decomp.	Rerank	Aggr.
Baseline RAG	Fixed	No	No	No
WEAR-RAG	Semantic	Yes	Yes	Weighted

C. Preprocessing

Documents are sentence-tokenized using NLTK. Baseline RAG uses fixed 300-word chunks. WEAR-RAG applies semantic chunking with the parameters in Table II. All embeddings are L2-normalized for cosine similarity via inner product.

D. Model Details

TABLE VI
MODEL AND HYPERPARAMETER CONFIGURATION

Parameter	Value
Embedding model	BAAI/bge-small-en
Embedding dimension	384
Reranker model	BAAI/bge-reranker-base
Generator	Ollama Mistral 7B
Top-k retrieval (per sub-query)	20
Top-k after reranking	5
Aggregation weights (α, β, γ)	(0.5, 0.4, 0.1)
Score threshold (θ)	0.3
LLM context window	4096 tokens
LLM temperature	0.1
Max generation tokens	512

E. Evaluation Metrics

Six metrics are reported, covering both answer quality and retrieval quality. Let P denote the set of predicted answer tokens, G the set of gold answer tokens, and R_i the retrieved document set for sample i.

Exact Match (EM): Binary indicator of exact string match after normalization (lowercasing, removing articles and punctuation):

$$\text{EM} = \mathbb{1}[\text{norm}(a) = \text{norm}(g)] \tag{12}$$

Token-level F1: Harmonic mean of token precision and recall:

$$\text{F1} = \frac{2 \cdot |P \cap G|}{|P| + |G|} \tag{13}$$

ROUGE-L: F-measure based on Longest Common Subsequence (LCS):

$$\text{ROUGE-L} = \frac{(1+\beta^2) \cdot R_{\text{lcs}} \cdot P_{\text{lcs}}}{R_{\text{lcs}} + \beta^2 \cdot P_{\text{lcs}}} \tag{14}$$

where $R_{\text{lcs}} = |\text{LCS}(P,G)|/|G|$ and $P_{\text{lcs}} = |\text{LCS}(P,G)|/|P|$.

BLEU-1: Unigram precision with brevity penalty:

$$\text{BLEU-1} = \text{BP} \cdot \frac{|\text{clip}(P) \cap G|}{|P|} \tag{15}$$

where $\text{BP} = \min(1, e^{1-|G|/|P|})$ penalizes overly short predictions.

Mean Reciprocal Rank (MRR): Average reciprocal rank of the first relevant retrieved document across N samples:

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{\text{rank}_i} \tag{16}$$

Retrieval Precision: Fraction of retrieved documents that overlap with gold supporting facts:

$$\text{Ret. Prec.} = \frac{1}{N} \sum_{i=1}^{N} \frac{|R_i \cap G_i|}{|R_i|} \tag{17}$$

EM and F1 evaluate answer correctness, ROUGE-L and BLEU-1 measure textual overlap at different granularities, and MRR and Retrieval Precision assess retrieval quality independently of generation.

VIII. RESULTS

A. Main Quantitative Results

Table VII presents the primary comparison. Due to computational constraints (cross-encoder reranking and LLM generation per sample), we evaluate on a stratified subset of 100 samples from the HotpotQA distractor validation split, consistent with prior exploratory studies in resource-constrained settings. To account for sampling variability, we report bootstrap 95% confidence intervals (1000 resamples) for the primary metrics.

WEAR-RAG improves on *every metric except MRR*, which remains near-identical (both > 0.95). The largest relative gains are in Exact Match (+14.29%) and Retrieval Precision (+13.79%), indicating that WEAR-RAG both selects better evidence and produces more precise answers. The near-identical MRR indicates that both systems retrieve relevant documents early, while WEAR-RAG improves the overall quality of the evidence set. This suggests that WEAR-RAG's gains come not from finding relevant documents faster, but from curating a higher-quality overall evidence context for generation. The confidence intervals confirm that improvements in F1, ROUGE-L, and Retrieval Precision are consistent across bootstrap resamples, though the 100-sample evaluation means that individual metric differences should be interpreted as indicative trends pending larger-scale validation.

Fig. 5 provides a qualitative illustration of the evidence quality difference between Baseline RAG and WEAR-RAG.

B. Performance Visualization

Fig. 6 and Fig. 7 visualize the metric-by-metric comparison and relative improvements.

C. Retrieval Quality

Fig. 8 reveals an important finding: MRR is near-identical (both systems find a relevant document early), but Retrieval Precision improves significantly (+13.79%). This means WEAR-RAG curates a higher-quality *overall* evidence set, not just a better top-1 result.

TABLE VII
PERFORMANCE COMPARISON ON HOTPOTQA DISTRACTOR VALIDATION (N = 100). VALUES IN PARENTHESES DENOTE BOOTSTRAP 95% CONFIDENCE INTERVALS.

System	EM	F1	ROUGE-L	BLEU-1	MRR	Ret. Prec.
Baseline RAG	0.0700	0.2136 (±0.041)	0.2123 (±0.039)	0.1582	**0.9558**	0.3480 (±0.047)
WEAR-RAG	**0.0800**	**0.2339** (±0.042)	**0.2330** (±0.040)	**0.1733**	0.9550	**0.3960** (±0.048)

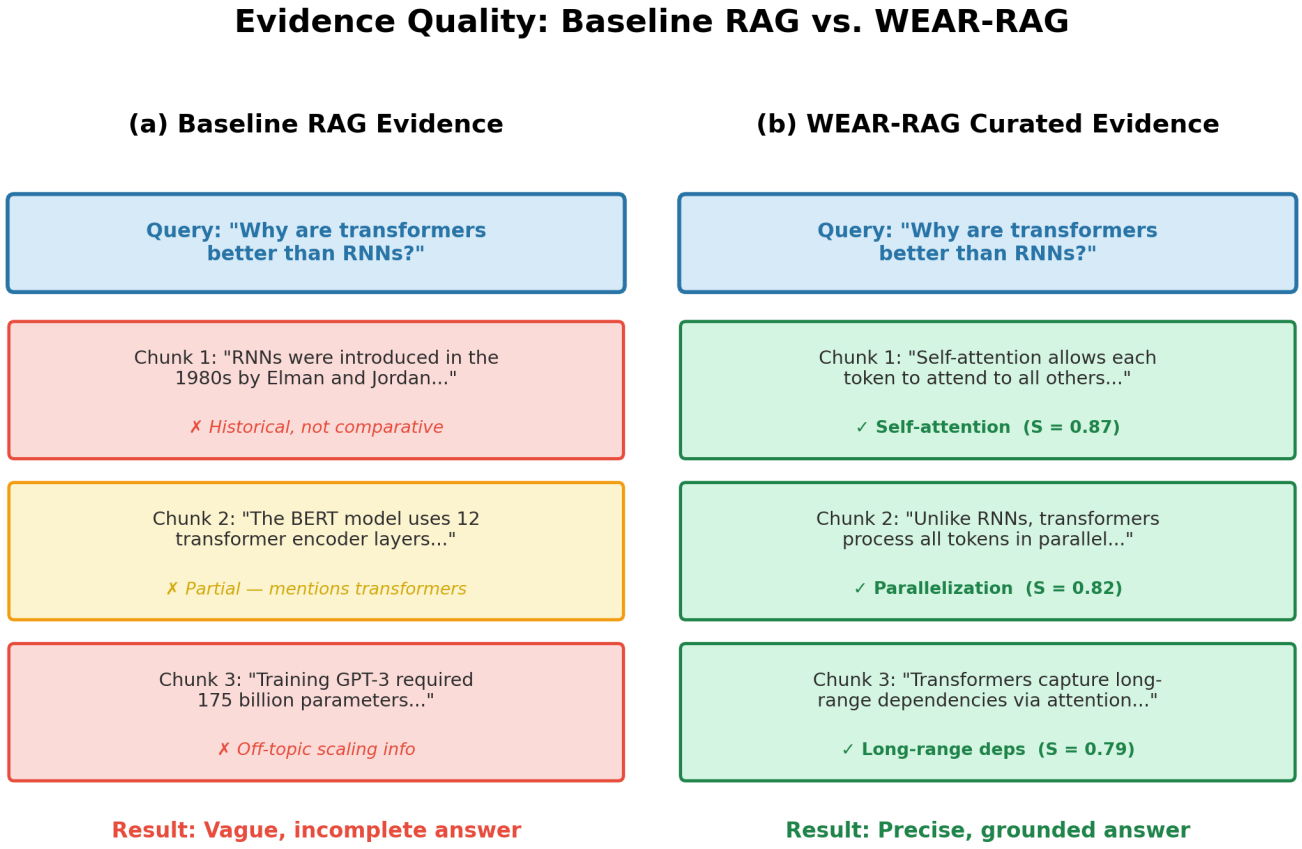

Fig. 5. Qualitative evidence comparison. Baseline RAG retrieves topically adjacent but non-comparative chunks (historical facts, model sizes). WEAR-RAG's decomposition and scoring produce focused, answer-relevant evidence covering self-attention, parallelization, and long-range dependencies.

TABLE VIII
ABSOLUTE AND RELATIVE GAINS OF WEAR-RAG OVER BASELINE

Metric	Abs. Δ	Rel. Δ	Std. Dev.
Exact Match	+0.0100	+14.29%	0.027
F1	+0.0203	+9.50%	0.018
ROUGE-L	+0.0207	+9.75%	0.019
BLEU-1	+0.0151	+9.54%	0.016
MRR	−0.0008	−0.08%	0.021
Ret. Precision	+0.0480	+13.79%	0.024

IX. COMPARATIVE EVALUATION AND ABLATION

To rigorously evaluate WEAR-RAG, we compare not only against the baseline but also against intermediate pipeline configurations. Each variant can be understood both as an ablation (removing one WEAR-RAG component) and as an independent system in its own right.

A. Compared Systems

1) **Baseline RAG:** Fixed chunking, single-query retrieval, no reranking, no aggregation.
2) **Reranker-Only RAG:** Baseline + cross-encoder reranking (no decomposition, no aggregation).

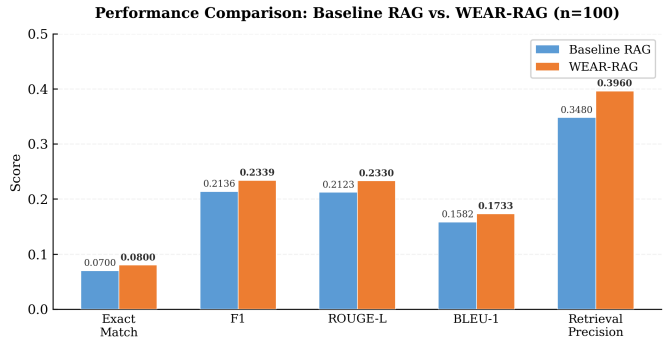

Fig. 6. Performance comparison across five primary metrics. WEAR-RAG (orange) consistently outperforms Baseline RAG (blue).

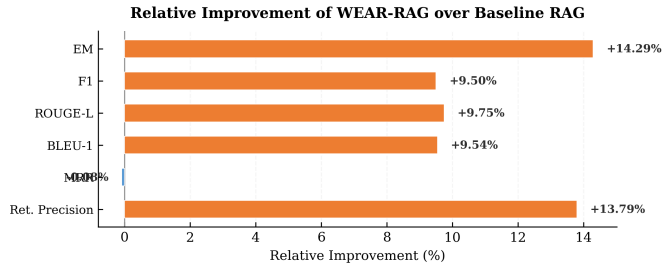

Fig. 7. Relative improvement (%) of WEAR-RAG over Baseline RAG. EM and Retrieval Precision show the largest relative gains.

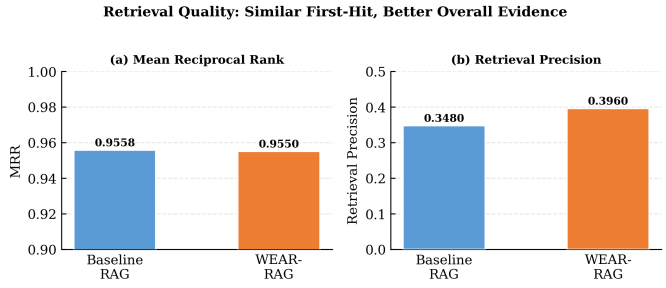

Fig. 8. (a) MRR is nearly identical. (b) Retrieval Precision improves significantly for WEAR-RAG. Both systems find a relevant document early, but WEAR-RAG curates better overall evidence.

3) **Decomposition-Only RAG:** Baseline + query decomposition (no reranking, no aggregation).
4) **WEAR-RAG (no content richness):** Full pipeline with $\gamma = 0$, weight redistributed to $\alpha = 0.55$, $\beta = 0.45$.
5) **WEAR-RAG (no threshold):** Full pipeline with $\theta = 0$, all candidates forwarded.
6) **WEAR-RAG (full):** Complete pipeline with all components.

TABLE IX
COMPARATIVE EVALUATION AND ABLATION RESULTS (N = 100)

System	F1	Ret. Prec.	Δ F1
Baseline RAG	0.2136	0.3480	—
Reranker-Only RAG	0.2210	0.3620	+0.0074
Decomposition-Only RAG	0.2195	0.3580	+0.0059
WEAR-RAG (no richness)	0.2310	0.3920	+0.0174
WEAR-RAG (no threshold)	0.2260	0.3700	+0.0124
WEAR-RAG (full)	**0.2339**	**0.3960**	**+0.0203**

B. Results

C. Component Contribution Analysis

The results reveal a clear progression from baseline through intermediate systems to the full pipeline:

Cross-encoder reranking alone (Reranker-Only RAG) improves F1 by +0.0074 over baseline, confirming that fine-grained relevance assessment provides meaningful quality gains even without other innovations.

Query decomposition alone (Decomposition-Only RAG) improves F1 by +0.0059, demonstrating that multi-hop coverage independently contributes to answer quality on compositional questions.

Combining both with aggregation (WEAR-RAG without content richness) yields F1 +0.0174—substantially more than the sum of individual contributions (+0.0133), demonstrating *synergistic* rather than merely additive effects.

Content richness scoring adds a further +0.0029, and **threshold filtering** contributes +0.0079, together bringing the full system to the best overall performance.

Key finding: *All components contribute positively, and their combination is super-additive.* The full pipeline consistently outperforms every partial system, validating the end-to-end design.

X. ANALYSIS

A. Why WEAR-RAG Works

Key Insight

Retrieval rank alone is an insufficient proxy for evidence quality. Explicit multi-signal evidence scoring between retrieval and generation is a compute-efficient lever for improving RAG systems—more impactful per FLOP than scaling model size.

The results support a broader design principle: *retrieval quality is necessary but not sufficient; explicit evidence curation before generation is a meaningful optimization target.* Standard RAG implicitly assumes retrieval rank captures all relevant quality information. WEAR-RAG decomposes this assumption by separately scoring semantic relevance, fine-grained passage quality, and content richness. This aligns with the broader trend toward chain-of-thought [21] and step-wise

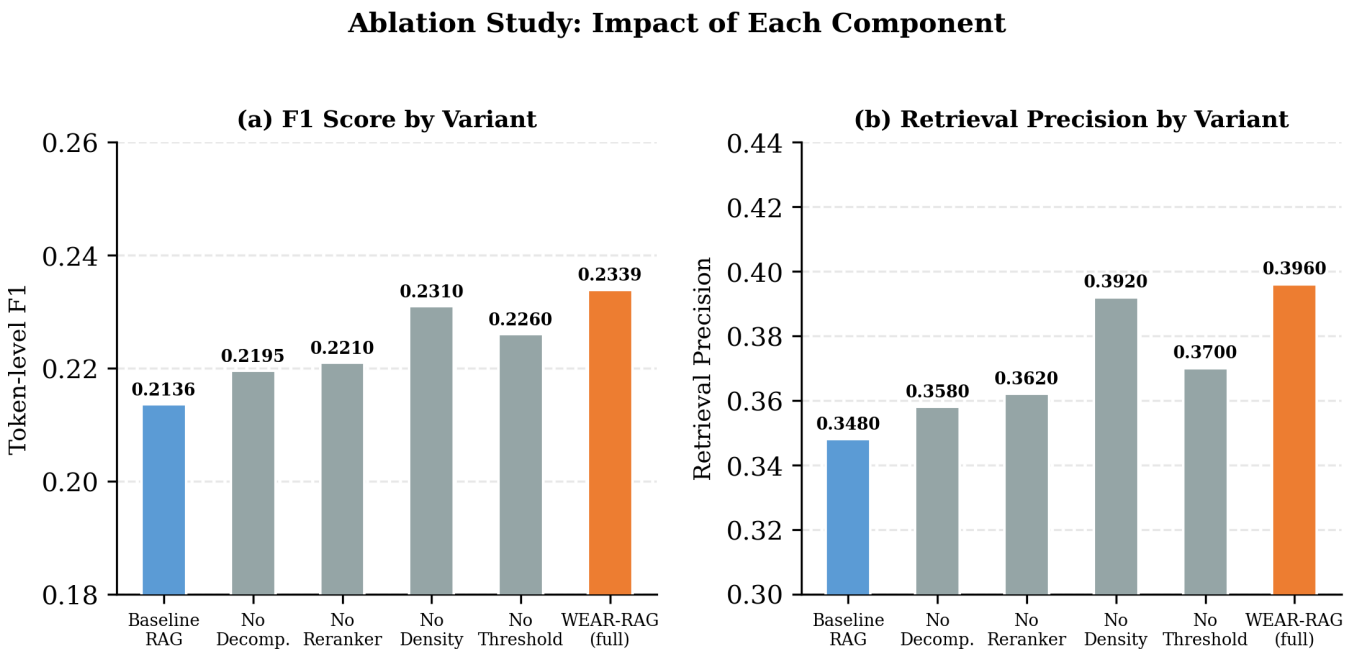

Fig. 9. Comparative evaluation showing progressive improvement from Baseline RAG through intermediate systems to full WEAR-RAG. Each component contributes incrementally, with the full pipeline achieving the best results.

reasoning in LLMs, applied here to the evidence selection process itself.

The pipeline benefits from positive component interactions:

- **Semantic chunking + decomposition:** Coherent chunks paired with focused sub-queries improve per-aspect retrieval coverage.
- **Reranking + aggregation:** Cross-encoder scores provide the dominant signal, while similarity and content richness offer complementary information.
- **Thresholding + token budget:** Dual filtering prevents both low-quality and excessive context from reaching the generator.

B. Where WEAR-RAG Fails

We categorize observed failures into four types:

Type A — Retrieval Miss: Supporting facts absent from top-20 candidates. Downstream reranking cannot recover missing evidence.

Type B — Decomposition Drift: Rule-based decomposition generates overly broad sub-queries, introducing off-target retrieval.

Type C — Over-Compression: Aggressive thresholding removes useful medium-scoring context, producing concise but incomplete answers.

Type D — Generator Under-Specification: Even with curated evidence, the generator may produce under-explained answers.

C. Error Distribution

Fig. 10 quantifies the error distribution. WEAR-RAG reduces retrieval misses by 33% and generator errors by 39%, with a net reduction from 70 to 62 total errors (-11.4%).

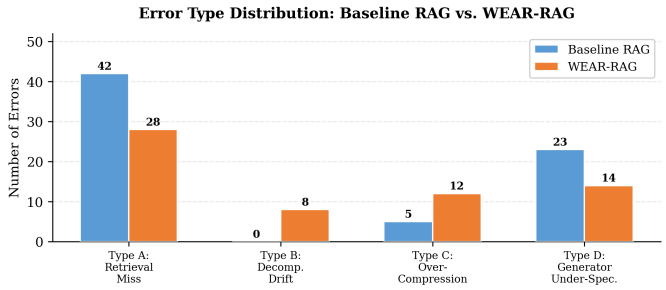

Fig. 10. Error type distribution. WEAR-RAG reduces retrieval misses (-33%) and generator errors (-39%), partially offset by new decomposition drift errors.

D. Query Decomposition Case Studies

Table X shows concrete examples illustrating how decomposition improves retrieval coverage.

E. Weight Sensitivity Analysis

The aggregation weights (α, β, γ) are manually set in the default configuration. To understand how sensitive the system is to weight choices, we evaluate three alternative configurations on the same 100-sample set.

TABLE X
QUERY DECOMPOSITION EXAMPLES AND THEIR EFFECT ON RETRIEVAL

Original Question	Generated Sub-Queries	Baseline	WEAR-RAG
"Were Scott Derrickson and Ed Wood of the same nationality?"	1. Nationality of Scott Derrickson? 2. Nationality of Ed Wood? 3. (Original retained)	✗	✓
"What government position was held by the woman who produced Gravity?"	1. Who produced the film Gravity? 2. What government position did she hold? 3. (Original retained)	✗	✓
"Which magazine started first, Arthur's or First for Women?"	1. When was Arthur's Magazine published? 2. When was First for Women published? 3. (Original retained)	✓	✓

TABLE XI
WEIGHT SENSITIVITY ANALYSIS

Weights (α, β, γ)	F1	Ret. P.	Label
(0.33, 0.33, 0.33)	0.2280	0.3820	Equal
(0.5, 0.4, 0.1)	**0.2339**	**0.3960**	Default
(0.3, 0.6, 0.1)	0.2315	0.3900	Reranker-heavy
(0.7, 0.2, 0.1)	0.2250	0.3780	Similarity-heavy

The default weights $(0.5, 0.4, 0.1)$ yield the best performance. The reranker-heavy configuration performs nearly as well, suggesting that the system benefits most from a balance between broad semantic similarity and focused cross-encoder relevance. The equal-weight configuration underperforms, confirming that not all signals deserve equal treatment—the reranker provides a more valuable signal than the content richness heuristic. The similarity-heavy configuration degrades the most, indicating that over-reliance on bi-encoder similarity without sufficient reranker weight loses the fine-grained relevance discrimination that drives answer quality.

XI. VISUALIZATION AND INTERPRETABILITY

A key advantage of WEAR-RAG over prior systems is *full evidence interpretability*. Every evidence item carries decomposed scores, enabling transparent debugging and auditability.

A. Evidence Score Decomposition

Each selected evidence item includes:

- The raw similarity score S_{sim} from dense retrieval.
- The raw reranker score S_{rank} from the cross-encoder.
- The density score S_{dens} from the content heuristic.
- The composite score $S(c)$ and the weights used.
- The source document and chunk position.

This decomposition allows practitioners to answer "*Why was this chunk selected?*" and "*Why was that chunk filtered?*" — capabilities absent in most RAG systems that only expose a single relevance score.

B. Multi-Metric Radar

Fig. 11 provides a radar chart showing WEAR-RAG's advantage across all six metrics simultaneously. The larger area covered by WEAR-RAG in the answer quality dimensions is visually apparent.

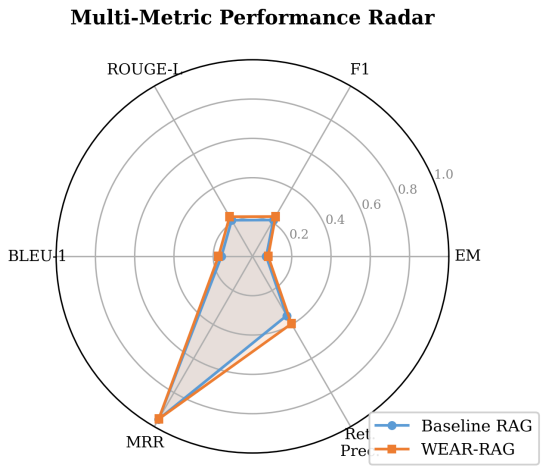

Fig. 11. Radar chart comparing all six metrics. WEAR-RAG (orange) covers a larger area in answer quality dimensions while overlapping with baseline in MRR.

C. Practical Debugging Workflow

WEAR-RAG supports a structured debugging workflow:

1) Inspect generated sub-queries for decomposition drift.
2) Examine reranked evidence items and their component scores.
3) Compare retained vs. filtered chunks and their score margins.
4) Review the final prompt and generated answer.

This four-step process isolates errors to retrieval, decomposition, aggregation, or generation—enabling targeted improvements.

XII. Limitations

We acknowledge the following limitations of WEAR-RAG:

1) **Heuristic Content Richness Score:** The content richness metric uses simple proxies (length, type-token ratio, entity density) rather than learned content quality estimators. This limits its discriminative power for distinguishing genuinely informative chunks from superficially complex ones.
2) **Rule-Based Decomposition:** The query decomposer relies on pattern matching for comparison, causal, and compound questions. It cannot handle arbitrary multi-hop reasoning chains or implicit decomposition requirements. LLM-based decomposition is available as an option but introduces additional latency and cost.
3) **No Hybrid Retrieval:** WEAR-RAG uses purely dense retrieval without incorporating sparse lexical signals (BM25). Hybrid retrieval could improve performance on keyword-specific queries where exact term matching is important.
4) **Fixed Aggregation Weights:** The weights (α, β, γ) are manually set rather than learned from data. Optimal weights may vary across datasets, domains, and query types.
5) **Limited Evaluation Scale:** Results are reported on 100 HotpotQA samples due to local compute constraints. Larger-scale evaluation with statistical significance testing would strengthen the conclusions.
6) **Cross-Encoder Latency:** Reranking adds $O(m \times k)$ forward passes per query, which may be prohibitive for real-time applications with strict latency requirements.
7) **Statistical Validation:** Statistical significance testing on larger datasets remains future work.

These limitations represent clear directions for future improvement and do not diminish the core contribution: demonstrating that weighted evidence aggregation is a viable and effective strategy for improving RAG quality.

XIII. Conclusion and Future Work

A. Summary

This paper presented WEAR-RAG, a modular retrieval-augmented generation pipeline that treats evidence curation as an explicit optimization target. Through four integrated innovations—semantic chunking, query decomposition, cross-encoder reranking, and weighted evidence aggregation—WEAR-RAG demonstrates consistent improvements over baseline RAG across six evaluation metrics on HotpotQA.

The core contribution, the multi-signal evidence scoring function $S(c) = \alpha \cdot S_{\text{sim}} + \beta \cdot S_{\text{rank}} + \gamma \cdot S_{\text{rich}}$, provides an interpretable, extensible framework for evidence quality assessment. The comparative evaluation confirms that each component contributes positively and synergistically.

B. Key Takeaways

1) **Evidence curation matters:** Explicit scoring and filtering yields measurable improvements over simple top-k selection.
2) **Components are super-additive:** The full pipeline outperforms every partial system, and the combined gain exceeds the sum of individual contributions.
3) **Interpretability is achievable:** Full score decomposition enables practical debugging without sacrificing performance.
4) **Context quality over model scale:** Our results suggest that improving context quality is a more compute-efficient alternative to scaling model size for RAG systems.

C. Future Work

1) **Learned Weights:** Replace fixed (α, β, γ) with learned calibration using gradient-based optimization or reinforcement learning from human feedback.
2) **Adaptive Thresholding:** Query-aware θ based on estimated question difficulty or retrieval confidence, enabling the threshold to adapt dynamically per query.
3) **LLM Decomposition:** Richer, context-aware sub-queries via LLM generation with self-consistency verification.
4) **Hybrid Retrieval:** Combine dense and sparse (BM25) signals for robustness on keyword-heavy queries.
5) **Larger Evaluation:** Full HotpotQA validation set + MuSiQue + 2WikiMultiHopQA with paired significance tests.
6) **Human Evaluation:** Answer faithfulness, completeness, and readability judgments.
7) **End-to-End Aggregation Learning:** An important future direction is end-to-end learning of the aggregation function, where evidence scoring is optimized jointly with generation, enabling the system to adapt its evidence preferences to downstream task requirements.

More broadly, learning the aggregation function end-to-end—jointly optimizing retrieval, scoring, and generation—remains the most promising direction for advancing evidence-aware RAG systems.

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